

and does affect performance during benchmarking. There's a flaw with the layout of this board—the 24-pin power connector is placed bang in the middle of the motherboard, and right between the CPU and graphics card—two very hot components; we have to wonder what BioStar was thinking!

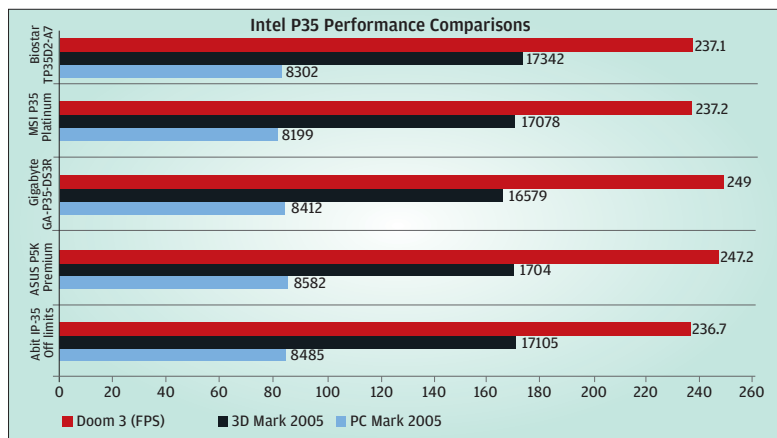
In terms of overclocking features, all these boards allow you to play around, but ASUS' BIOS is very newbie-friendly—and you can safely over-volt your CPU with minuscule increments of 0.00625 volts!

ECS' P35 doesn't support memory frequencies above 800 MHz—a pity for those who happen to have fast memory—and hence limits overclocking. Additionally, the BIOS doesn't allow you to fix the multiplier and there's no option to enable or disable EIST (Enhanced Intel SpeedStep—a standard option on boards for Core 2 Duo processors).

MSI's P35 Neo is a nice value offering at Rs 6,600 after all the Rs 12,000-plus offerings from ASUS and Gigabyte. It's well laid out, though shy on some of the extra frills like E-SATA, POST displays and power buttons on the board, copper heatpipes etc.

Performance

We noticed that the DDR3 motherboards performed slightly better—this shows that the Intel Core 2 Duo is sensitive to bandwidth



changes (800 MHz vs. 1066 MHz) in spite of the slower timings that plague DDR3.

ASUS' Blitz Formula, P5K Premium, and Gigabyte's GA-P35 DS3R were the best performers, followed closely by ASUS' P5K Deluxe and Abit's IP-35 Pro and MSI's P35 Platinum. We aren't really looking at the DDR3 based solutions as viable till the cost of DDR3 memory falls off its current lofty perch.

Incidentally, the Abit IP-35 Pro cheats a little by keeping the FSB at a default of 272 MHz, and you must manually get it down to 266 MHz (which should be the default speed)—this does

How We Tested

We categorised our motherboards based on the way customers like to buy them—on the basis of the all important chipset:

- Intel P35-based
- NVIDIA 680i / 650i-based
- Intel P965-based
- All boards with integrated graphics (for those who can't be bothered with an add-on graphics card)

We used the following test components:

	Intel	AMD
Processor	Core 2 duo X6800 @ 2.93 GHz	Athlon X2 6400+ @ 3.2 GHz
Memory	Corsair Dominator PC2 8500 @ 800 MHz (4-4-4-12) / Kingston PC3 8500 @ 1066 MHz (7-7-7-24)	Corsair Dominator PC2 8500 @ 800 MHz (4-4-4-12)
Graphics	NVIDIA 8800 GTX @ 575/1800 MHz	NVIDIA 8800 GTX @ 575/1800 MHz
HDD	Western Digital Raptor 10,000 rpm 74 GB x 2	Western Digital Raptor 10,000 rpm 74 GB x 2
OS	Windows XP Prof SP2	Windows XP Prof SP2
VGA Driver version	Forceware 158.22	Forceware 158.22

In addition, we installed the latest BIOS updates and drivers, downloaded from the manufacturers' sites.

Features

Besides all the regular features, we also took a special look at the layout of the motherboards, and rated them on general component layout—capacitors around the CPU region, placements of IDE and SATA connectors, power connectors, memory slot placement, the type of VGA slot retention mechanism and so on. Due to space restrictions in these pages, we've shown an average score of all these parameters.

We also looked very carefully at overclocking options that many motherboards today provide. With overclocking becoming very idiot-proof, anyone can play around with BIOS settings safely. Enthusiasts will also find a lot of valuable information as far as CPU and memory voltage parameters go.

We haven't talked about RAID support—all boards today support RAID 0, 1, 0+1, 5 and 10, so this data is redundant and unnecessary.

Performance Tests

Synthetic Tests

PC Mark 2005: This benchmark runs 48 discrete tests to test each performance-related subsystem—the processor, memory, video and storage. Finally, it throws out a final score—which is a weighted average of the individual results.

SiSoft Sandra 2007 Engineer: Sandra 2007 has specific benchmarks to test all the subsystems of a PC. It tests a CPU's floating point crunching ability under both arithmetic and multimedia applications. The application supports multiple cores and scales well in this regard. It also tests the read and write performance of the storage subsystem, and the memory's bandwidth.

3D Mark 2005: Our graphics benchmark of choice, mainly because it allows for greater variation in results than the newer 3D Mark 2006.

Real World Tests

Video encoding: This is a CPU-intensive test and tells us exactly how optimised the motherboard in question is.

File transfer test: We copy a single 4 GB file from one physical hard disk to another. This tells us a lot about Southbridge data path optimisations on each motherboard, and how fast data is moving around between hard disks.

Doom 3 And Splinter Cell Chaos Theory: *Doom 3* uses a very sensitive engine that even responds to changes in memory timings or a hard drive change! *Chaos Theory* is very scalable and very CPU- and shader-intensive. At a resolution of 640x480 with all details off, we're basically allowing the CPU to run wild while limiting the role of the graphics card. At 1600x1200—the maximum settings—any game becomes CPU-limited and any increase in frame-rate can be attributed to the graphics card and its respective data paths on the motherboard.